

InventOR: A Human-Reviewed, Prompt-Configured Large Language Model Decision Support System for Material-Level Inventory Control

Aris Dressino^{a,*}

^a*TUM School of Management, Technical University of Munich, Arcisstraße
21, Munich, 80333, Germany*

Abstract

Material planners must set a reorder point, a safety stock and an order quantity for thousands of plant-material pairs, usually under ERP parameters that were last revised long ago. We define and evaluate *InventOR*, a decision support system (DSS) for this recurring decision. For each material, a large language model (LLM) configured only through a prompt, without training or fine-tuning, proposes a reorder-point/order-quantity (r, Q) policy with a written justification and an escalation flag. Deterministic software gates the proposal against minimum order quantities and storage limits and replays it on held-out demand before the planner, who remains the decision-maker, reviews it. We derive six design requirements and evaluate the artifact on 365 plant-material pairs from three plants of an industrial manufacturer, using three independent LLM runs, seven reference arms and an ablation that withholds the ERP planning parameters. In a hard-limit storage replay, the LLM policies cost about 11% less than ERP-derived parameters at a mean fill rate 0.1–0.4 points lower, but 6%, 8% and 18% more than

*Corresponding author. Telephone: +49 89 289 25000. ORCID: 0009-0003-0693-9051.
Email address: `aris.dressino@tum.de` (Aris Dressino)

history-only, Croston/SBA and empirical-quantile (r, Q) baselines. The saving comes from carrying 13–17% less inventory with 11–20% more stockout days; when shortages are priced at one unit cost, ERP-derived parameters are cheaper and a simulation-tuned policy is cheapest. Static storage checks are insufficient: simulated trajectories exceed storage limits for 53–66% of pairs. InventOR is a checkable, human-reviewed candidate generator, not an optimizer that beats data-driven baselines; benefits to planners are untested. We contribute design principles, review rules and a research agenda.

Keywords: Decision Support Systems, Large Language Models, Generative Artificial Intelligence, Inventory Control, Human–AI Decision Making, Design Science, Operations Research

1. Introduction

Replenishment parameters in manufacturing are set per plant-material pair: a reorder point, a safety stock and an order quantity. In large ERP systems these values often remain at levels set years earlier, while demand, supplier performance and storage constraints change. Revisiting them requires analysts who can fit demand and lead-time statistics, apply inventory theory, respect minimum order quantities (MOQs) and storage limits, and explain the result to planners. A conversational large language model (LLM) with a code interpreter appears to offer this scarce skill, and a growing literature explores such tools for supply-chain and operations decisions [1, 2, 3, 4, 5].

Three problems limit its use. First, LLM outputs can be unfaithful or fabricated [6], so a proposed policy is a candidate, not a decision. Second, superiority claims over ERP parameters are hard to interpret when the LLM and the comparators receive different information. Third, a storage rule that

15 holds when a policy is written down can be violated as stock builds up over
16 time.

17 This paper treats the problem as one of decision support system (DSS)
18 design. We define *InventOR* as a human-reviewed, model-driven DSS for
19 the recurring decision of setting replenishment parameters per plant-material
20 pair (Section 2). A planner remains the decision-maker. The system supplies,
21 for each material, a candidate (r, Q) policy proposed by an enterprise LLM
22 application that is configured only through a prompt and receives no project-
23 specific training or fine-tuning, together with a justification, an escalation
24 flag and a counterfactual simulation; deterministic software parses, gates
25 and simulates every proposal before the planner reviews it and alone decides
26 whether any ERP parameter is changed.

27 We follow a design-science logic [7, 8]: we state problem and scope, derive
28 design requirements, build the artifact and evaluate it on real industrial
29 data. Four research questions structure the paper: how an LLM should be
30 embedded so that its proposals are bounded, verifiable, reproducible and
31 reviewable (RQ1); what cost and service they achieve when the information
32 available to each approach is controlled (RQ2); whether they respect storage
33 limits along simulated trajectories and are stable across runs (RQ3); and
34 when a planner should trust, escalate or reject a proposal (RQ4).

35 Our contributions are: (C1) a definition and scope of the DSS, with six
36 design requirements that make the role of the LLM explicit and bounded
37 (Section 2); (C2) an architecture and operating rules that separate gener-
38 ation, deterministic gating, simulation and planner review, with trust, es-
39 calate and reject rules and design principles drawn from the application
40 (Sections 4 and 8); (C3) an information-controlled, trajectory-level evalua-
41 tion with seven reference arms, an LLM ablation denied the ERP planning

parameters and a hard-limit storage replay as the primary result (Section 5);
and (C4) complete-cohort industrial evidence from three LLM runs on 365
plant-material pairs, with managerial and research implications.

Our main findings are as follows. Against ERP-derived parameters the
LLM policies are about 11% cheaper at a mean fill rate 0.1–0.4 points lower,
but the LLM costs 6%, 8% and 18% more than history-only, Croston/SBA
and empirical-quantile (r, Q) baselines at essentially equal service, and its
saving over ERP comes from carrying less inventory at the price of more
stockout days. Storage limits that hold in a static check are violated along
unconstrained trajectories for 53–66% of pairs. The simulation therefore
does not show superior optimization by the LLM; it shows that a prompt-
configured LLM can produce checkable candidate policies without code or
training, whose benefit to planners remains to be tested.

2. Problem, Scope and Definition

2.1. The decision problem

The decision supported is the periodic (re)setting of a reorder point, a
safety stock and an order quantity for each plant-material pair held in an
ERP system. In the classification of Gorry and Scott Morton [9] it is a partly
structured operational-control decision: the calculation is textbook, but the
choice of estimator, the treatment of intermittent or trending demand, the
plausibility of recorded lead times and the handling of constraints require
judgement that is applied inconsistently when thousands of materials com-
pete for a few analysts' time. The decision-maker is the material planner;
hard constraints are the minimum order quantity (MOQ) and a physical stor-
age limit; and feedback is slow, because a changed parameter shows its effect

only in later demand. The scope is one pair at a time (single item, single echelon, continuous review, lost sales, (r, Q) family); order release, supplier selection, forecasting-model development and multi-echelon networks are out of scope, and the LLM is a candidate generator that does not forecast, place orders, simulate or decide.

2.2. Definition and level of automation

Definition. *InventOR is a human-reviewed, model-driven decision support system that, for one plant-material pair at a time, (i) obtains from a large language model, configured only by a written prompt, a candidate (r, Q) policy with a justification and an escalation flag; (ii) verifies the candidate deterministically against explicit constraints; (iii) evaluates it by counterfactual simulation against reference policies on held-out demand; and (iv) presents the result to a planner, who alone decides whether ERP parameters are changed.*

In the classical DSS architecture of data, model and dialog components [10, 11], the LLM and the simulator form the model component, the material-scoped data extract and the prompt form the data and knowledge components, and the review interface is the dialog component (Section 4.1). In terms of the levels of automation of Parasuraman et al. [12], InventOR operates at a low-to-intermediate level: it selects and suggests one alternative and leaves execution to the human. Accepting automated suggestions invites complacency and automation bias [13], which the review rules of Section 4.3 are meant to counter.

91 2.3. Design requirements

92 The requirements follow from the known unreliability of LLM outputs
 93 [6], guidance on human–AI interaction and accountability [14, 15], run-to-
 94 run nondeterminism of LLM inference [16] and the constraints of inventory
 95 control. Table 1 ties each to a design feature (Section 4) and an evaluation
 96 (Section 6).

Table 1: Design requirements, the design features that address them and where they are evaluated.

ID	Requirement	Design feature	Evaluation
DR1	Bounded generation: the LLM proposes and has no authority	Narrow candidate-generator role; prohibited methods; no order placement	Sections 6.1, 6.3
DR2	Verifiable feasibility along time, not only at a point	Deterministic gate; hard-limit trajectory replay	Sections 6.1, 6.5
DR3	Counterfactual evidence for the reviewer	Held-out simulation against reference arms	Sections 6.2, 6.8
DR4	Information transparency: comparisons control what each arm sees	Seven reference arms; LLM ablation without ERP parameters	Sections 6.3, 6.4
DR5	Reproducibility and accountability of the configuration	Frozen prompt file; per-call log; repeated runs	Section 6.7
DR6	Human control with explicit escalation	Trust, escalate and reject rules; escalation flag	Sections 4.3, 6.6

97 3. Related Work and Positioning

98 *Decision support systems and human–AI decision making.* DSS research
 99 has long distinguished data-, model- and knowledge-driven systems and has
 100 stressed that the value of a DSS lies in improving the judgement of a hu-
 101 man decision-maker rather than in replacing it [9, 10, 11]. Research on
 102 human–automation interaction shows that the way suggestions are scoped,
 103 presented and verified determines whether users rely on them appropriately
 104 or over-rely on them [12, 13],. Recent DSS work derives design principles
 105 for LLM-based assistants [17], reports design and pilot implementations of
 106 AI-enabled decision support in operations [18], and shows that the uptake of

107 algorithmic advice depends on transparency and on the user’s metacognition
108 [19, 20]; evaluations of data-driven forecasting systems in practice likewise
109 stress lessons beyond accuracy [21]. Model-driven support also shapes how
110 groups handle disagreement [22], and human override of algorithmic safety-
111 stock advice is a documented risk [23]; InventOR therefore keeps the planner
112 as decision-maker and makes the evidence for each proposal explicit.

113 *LLMs in operations and supply-chain decisions..* Reviews and frameworks
114 map the opportunities for generative AI in supply chains [1, 2, 24, 25, 3, 4, 26]
115 and stress that adoption depends on assigned human roles, data quality and
116 governance [5, 27, 28, 29, 30]. Evidence from manufacturing planning is thin:
117 a fine-tuned LLM for digital-manufacturing optimization [31], a transformer
118 trained to set order timing and quantity and benchmarked against (r, Q) [32],
119 and experiments showing that ChatGPT reproduces human decision biases
120 of managers [33]. Most technical LLM work translates problem statements
121 into optimization models and solver code [34, 35, 36, 37], studies multi-agent
122 control in inventory flows [38, 39], disruption modelling [40] or classical OR
123 tasks such as routing [41]. The closest LLM study to ours uses an LLM
124 as a forecasting copilot for spare-parts demand, combined with simulation,
125 and reports high variability for irregular demand that can be reduced by
126 averaging repeated queries and enriching prompts [42]. We test run-to-run
127 variability directly (Section 6.7); the difference is that our LLM proposes a
128 complete policy, not a forecast.

129 *Data-driven inventory control without LLMs..* Learning-based replen-
130 ishment is well established, including data-driven policies learned from
131 sequentially observed non-stationary demand [43] and forecasts augmented
132 with leading indicators [44], and heuristics remain competitive in constrained

133 settings such as perishable inventory [45]. Deep reinforcement learning has
 134 been applied to omnichannel and multi-echelon systems [46, 47, 48] and
 135 aerospace replenishment [49]; deep-learning forecasting has been integrated
 136 with replenishment [50]; contextual distributionally robust optimization
 137 uses side information to set order quantities [51]; and distribution-free
 138 online algorithms handle capacitated warehouses under uncertain price and
 139 demand [52]. The closest operational neighbour is a trained data-driven
 140 framework that jointly sets replenishment parameters, safety stock and
 141 safety lead time on real automotive data [53]. Reinforcement learning has
 142 also been combined with demand forecasting to optimize order points [54].
 143 These methods need training or tuned hyperparameters; InventOR needs
 144 none, at the price of a generator whose logic cannot be inspected beyond its
 145 written justification.

146 *Intermittent and spare-parts demand.* Material histories in manufacturing
 147 are frequently intermittent or lumpy. The statistical literature offers special-
 148 ized estimators and classification schemes [55, 56, 57, 58, 59], and recent stud-
 149 ies develop feature-based forecast combinations and temporal aggregation for
 150 intermittent demand [60, 61], forecasting for spare parts across product life-
 151 cycle phases [62], and integrated forecasting and planning or multi-echelon
 152 stocking models for repairable and spare parts [63, 64]. Manufacturers’ choice
 153 of safety buffers is examined in [65].

154 In summary, InventOR is not a new inventory model and does not claim
 155 to beat trained or optimized methods; it contributes a DSS design with a
 156 bounded LLM generator, an information-controlled evaluation and industrial
 157 evidence. The (r, Q) policy it uses is the standard workhorse of continuous-
 158 review control [66, 67, 68].

159 4. The InventOR Decision Support System

160 4.1. Architecture and components

161 Following the classical DSS components [10, 11], InventOR has a *data*
162 component (a material-scoped extract of daily demand, lead times, working-
163 day flags, cost and constraint parameters and ERP reference values; the
164 fields supplied are controlled and logged; DR4), a *knowledge* component (the
165 system prompt file defining the task, the prohibited methods, the storage
166 rules and the output schema, Supplementary File S1; DR1, DR5), a *model*
167 component split into a generator (an enterprise LLM application with a code
168 interpreter, configured only by the prompt, proposing one (r, Q) policy with
169 a justification and an escalation flag; DR1) and a deterministic verifier and
170 simulator (parser, gate, held-out simulation with hard storage-limit replay
171 and reference arms; DR2, DR3), a *dialog* component (policy, justification,
172 flag and simulated comparison shown to the planner under the trust, escalate
173 and reject rules; DR6) and a *governance* layer (frozen prompt file, per-call
174 log of input hash, time, session and raw response, repeated runs; DR5). The
175 generator is the only non-inspectable component. In the evaluated prototype
176 the dialog component is the per-pair output record; no interactive interface
177 was built or evaluated.

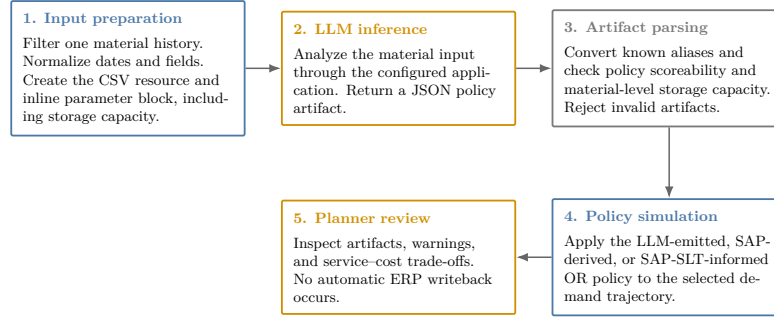
178 4.2. Operation in four steps

179 For one material, InventOR answers how much safety stock to hold, at
180 what level to reorder and how much to order, in four steps (Figure 1).

181 **Step 1. Generate.** The LLM, accessed through an enterprise application
182 with a code interpreter, receives one material’s history (daily de-
183 mand, realized lead times, working-day flags) and a short parameter

InventOR prototype workflow

Executed material-level path from prepared history to planner-reviewable simulation output.



Deterministic components reproduce simulation results from stored artifacts; LLM inference itself may vary.

Figure 1: InventOR workflow: material data are prepared, an LLM proposes a policy, deterministic software parses, checks and simulates it, and a planner reviews the result.

block (unit cost, holding rate, ordering cost, MOQ, service target, storage limit, and current ERP safety stock and safety lead time for comparison). A written prompt (the only configuration; no weights are changed) [69] instructs it to compute demand and lead-time statistics, apply an economic-order-quantity (EOQ) argument and a normal-approximation safety stock, respect the storage limit, and report a structured policy with a short justification and escalation flag.

Step 2. Gate. Deterministic code extracts the policy, rejects malformed or infeasible outputs (missing or non-numeric fields, non-positive reorder point or order quantity, negative safety stock, reorder point below safety stock, safety stock plus order quantity above the storage limit) and retries the generation up to three times. The backtest applies one further admission check, that the order quantity is not below the MOQ.

Step 3. Simulate. Each accepted policy is replayed on the material’s de-

mand observed *after* the data cutoff, a held-out evaluation in the sense of out-of-sample forecast testing [70, 71], together with the comparator policies, producing holding and ordering cost, fill rate and stockout days.

Step 4. Review. A planner sees the policy, the LLM’s justification and flag, and the simulated comparison. Only the planner changes ERP parameters.

The LLM is a *candidate generator* whose output is never trusted without the gate; it does not forecast, place orders or run the simulation. The prompt prohibits Monte Carlo simulation, distribution fitting, machine-learning libraries and specialized intermittent-demand estimators, so that the policy rests on transparent statistics.

4.3. Trust, review and reject rules

We propose three rules for planners, in line with guidelines for human–AI interaction and AI auditing [14, 15]. The rules and the 0.5-point threshold are design proposals not evaluated with planners; the trust rule would be met by a minority of pairs (the LLM is cheaper than the history-only OR in 74–77 of 346).

Trust (apply after review) when the proposal passes the gate, the simulated trajectory respects the storage limit when orders are truncated at it, and the simulated cost is not above that of the best history-only OR reference (Croston/SBA or empirical quantile) with a fill rate not more than about 0.5 points lower.

Escalate when the LLM raises an escalation flag, the material is lumpy or

224 has strongly seasonal or trending demand, the plant or calendar data were
225 repaired, or the policy differs strongly between repeated generations.

226 **Reject** when the proposal fails the gate after the retry budget, exceeds the
227 storage limit in the hard-limit replay, or is dominated (higher cost and lower
228 fill) by a history-only reference.

229 *4.4. Provenance and reproducibility of the configuration*

230 The system prompt, a plain-text file deployed unchanged in the enter-
231 prise application, defines the task, forbidden methods, capacity rules and
232 output schema (Supplementary File S1). Every call logs the SHA-256 of its
233 input, UTC time, session identifier and raw response. The platform does
234 not expose the model identifier or sampling settings; the evaluated model
235 (Anthropic Claude Sonnet 4.6 with a code interpreter) is operator-declared.
236 Reproducibility rests on the prompt file, the call logs and three independent
237 runs (Section 6.7).

238 **5. Application and Evaluation Design**

239 *5.1. Evaluation approach*

240 The evaluation addresses the design requirements (Table 1): validity and
241 feasibility (DR1, DR2), cost and service against reference arms on held-out
242 demand (DR3), the role of the information given to the LLM (DR4), storage
243 along trajectories (DR2), stability across runs (DR5) and escalation (DR6).
244 The primary basis is the hard-limit replay with working-day receipts and no
245 shortage penalty; sensitivity analyses vary these choices. Following design
246 science [7, 8], the evaluation is artificial and ex post in the framework of
247 Venable et al. [72]: real data and a faithful simulation but no live users. It

248 establishes technical efficacy, feasibility and reliability, not usability, adop-
249 tion or improved planner decisions.

250 5.2. Industrial application and data

251 The application is a pilot at a large manufacturer. The dataset contains
252 233,078 daily records for 411 plant-material pairs from three plants (A, B and
253 C), with a data cutoff on 10 March 2026 and a post-cutoff horizon observed
254 to 27 August 2026. Of the 411 pairs, 365 are eligible for scoring (234 in
255 Plant A, 65 in Plant B, 66 in Plant C) and are sent to the LLM. For 19
256 pairs (all in Plant A) the MOQ plus the reference safety stock exceeds the
257 storage limit, so no comparator can satisfy both rules; these are excluded
258 from outcome comparison, leaving a *common-feasible cohort* of 346 pairs
259 (215, 65 and 66). Demand is concentrated (the five largest materials account
260 for 40.3% of demand). The source working-day flag of Plant C is all zero; we
261 reconstruct it from the same-date majority of Plants A and B and test this
262 repair in Section 6.8. Data quality is a known threat to decision support [73]
263 and inventory records [74, 75, 76].

264 Each pair is sent as a material-scoped CSV with daily demand, recorded
265 lead times, a working-day flag, the minimum order quantity, unit cost,
266 holding-cost rate, ordering cost, service-level target, the storage limit, and
267 two *ERP planning parameters* (current safety stock and safety lead time),
268 which are withheld in the ablation arm. The CSV also carries supplier and
269 lane context fields that the prompt does not use.

270 5.3. Reference arms and information content

271 ERP safety stock and safety lead time embed planner judgement and are
272 alternative ways of buffering unreliability [77], and planned lead times can

drift upward [78]. The LLM and the classical comparators therefore do not necessarily use the same information, and we construct arms that make it explicit (Table 2).

Table 2: Comparison arms and the information each uses. All arms respect MOQ and are projected to the storage limit when possible.

Arm	Parameters from	ERP used	inputs	Purpose
SAP-derived	ERP safety stock; EOQ order quantity	Safety stock		ERP-derived reference (not observed practice)
SLT-informed OR	Stationary (r, Q) with ERP safety lead time added to the lead time	Safety lead time		OR policy using the ERP safety lead time
ERP-floor OR	SLT-informed OR with safety stock floored at the ERP value	Both		OR arm matched to the LLM's full information
History-only OR	Stationary (r, Q) from demand and lead-time history, mean lead time	None		ERP-independent baseline
Croston/SBA OR	(r, Q) with the Syntetos–Boylan-approximated Croston demand rate and its forecast-error spread, mean lead time	None		Intermittent-demand baseline
Empirical-quantile OR	(r, Q) with the reorder point set to the service-level quantile of historical rolling lead-time demand	None		Distribution-free baseline
Simulation-tuned (r, Q)	Reserve factor and order-quantity multiple chosen by simulation on the last 250 working days before the cutoff	None		Trained data-driven baseline
LLM (runs 1–3)	LLM proposal with all CSV fields (Section 5.2)	Both		Main treatment
LLM without ERP	LLM with ERP safety stock and safety lead time set to “not supplied”	None		Ablation: what the LLM does when it cannot anchor on ERP

Three further arms use no ERP input and represent stronger data-driven practice. The *Croston/SBA OR* replaces the mean daily demand by the Croston estimate with the Syntetos–Boylan bias correction [55, 56] (smoothing constant 0.1), uses the root mean squared one-step forecast error as demand spread and the history-only lead-time moments, which suits inter-

281 mittent demand [57]. The *empirical-quantile OR* sets the reorder point to the
 282 service-level quantile of historical rolling lead-time demand over the rounded
 283 mean lead time. The *simulation-tuned* (r, Q) policy selects, from a grid of
 284 reserve factors and order-quantity multiples around the history-only policy,
 285 the cheapest candidate whose fill rate on the last 250 working days *before*
 286 *the cutoff* meets the service target (or the best-fill candidate if none does).
 287 No post-cutoff data are used in any of these arms. We do not implement a
 288 deep reinforcement-learning baseline [47, 46]; this is a limitation (Section 8).

289 5.4. Simulation, capacity replay and metrics

290 Each pair is simulated independently on the post-cutoff horizon as a
 291 single-item, single-echelon continuous-review (r, Q) system [66, 79, 80, 68, 67]
 292 with lost sales [81], in the spirit of simulation-based experiments on supply
 293 and timing uncertainty [82], a shared opening stock, an empty pipeline and
 294 arrivals after the rounded mean lead time in working days (Appendix Ap-
 295 pendix A). MOQs are treated as hard order-size constraints, as imposed
 296 by suppliers [83], and storage limits as capacity bounds on the replenish-
 297 ment decision [52]. The primary metrics are the mean material-horizon
 298 holding-plus-ordering cost, the mean material-level fill rate, the aggregate
 299 (volume-weighted) fill rate and the number of stockout days. We report
 300 paired differences with 10,000-sample bootstrap 95% intervals.

301 A static check $SS + Q \leq \text{max_storage_units}$ does not show that storage
 302 limits hold as inventory evolves, because the reorder point includes lead-time
 303 demand and inventory on order can exceed the limit on arrival. We therefore
 304 record the post-receipt on-hand stock for every policy and day and report
 305 the share of pairs and days above the limit (the *unconstrained* simulation),
 306 and we *replay* every policy with each order truncated to the storage limit

307 minus the current inventory position (the *hard-limit replay*), which is our
308 primary evaluation.

309 5.5. Repeated runs and sensitivities

310 To examine run-to-run variation, the entire cohort is generated three
311 times (runs 1, 2 and 3) with the same prompt file and the same data. Run 1
312 serves as the reference for tabulated results unless stated otherwise; runs 2
313 and 3 provide replicates. Sensitivity analyses vary the working-day recon-
314 struction (majority, any-plant, strict-all, exclusion of Plant C and calendar-
315 day arrivals), the shortage penalty (zero, one and five times unit cost).

316 6. Results

317 6.1. Completion and policy characteristics

318 All four runs (three with the full input and one ablation, Section 6.4)
319 return a valid, storage-feasible (r, Q) policy for all 365 pairs after at most
320 three attempts per pair. The validity gate rejects non-positive reorder points
321 and order quantities, negative safety stocks and reorder points below the
322 safety stock. The ERP safety stock is zero for 198 of the 365 pairs (188 of
323 the 346 in the cohort below), so for a majority of pairs the SAP-derived arm
324 holds no safety stock at all.

325 6.2. Headline comparison with ERP-derived parameters

326 Unless stated otherwise, all results use the hard-limit replay on the
327 346-pair common-feasible cohort with working-day receipts and no shortage
328 penalty (Section 6.5 gives the unconstrained simulation). Table 3 compares
329 all arms; its last columns give the LLM cost relative to each arm in the three
330 runs (positive: the LLM costs more). In run 1 the LLM policies cost 11.4%

less than SAP-derived policies and 30.5% less than the SLT-informed OR (mean 2,928 versus 3,305 and 4,211), at a mean fill 0.18 points lower than SAP-derived (98.11% versus 98.30%; interval -0.54 to $+0.10$). The mean cost difference against SAP-derived is -377 (95% interval -651 to -154); the LLM is cheaper in 241 of 346 pairs and, in 226, simultaneously no more expensive and no worse in fill. It incurs more stockout days (910 versus 804).

Relative to SAP-derived parameters the LLM carries about 17% less average inventory (16,256 versus 19,565 units) and about 16% lower holding cost at about the same number of orders per year (17.3 versus 17.6), and it accepts 11–20% more stockout days (runs 3 and 2). The saving is therefore a trade of inventory for service, not a better ordering rhythm, and Section 6.8 shows that once shortages carry a cost this trade no longer favours the LLM.

Table 3: Hard-limit replay, common-feasible cohort ($n = 346$), working-day receipts, zero shortage penalty. Cost is the mean material-horizon holding-plus-ordering cost; average inventory is in units; orders are per year. The last column gives the LLM cost relative to the arm in runs 1/2/3 (positive: the LLM costs more).

Arm	Mean cost	Holding	Ordering	Avg. inv.	Orders	Mean fill (%)	Aggr. fill (%)	Stockout days	LLM vs arm (%)
ERP-derived (SAP)	3,305	2,172	1,132	19,565	17.6	98.30	79.34	804	$-11.4 / -11.3 / -11.4$
SLT-informed OR	4,211	2,608	1,603	21,614	24.5	98.23	79.15	831	$-30.5 / -30.4 / -30.5$
ERP-floor OR	4,248	2,633	1,616	21,937	24.7	98.23	79.15	831	$-31.1 / -31.0 / -31.1$
History-only OR	2,764	1,725	1,039	15,972	16.3	98.14	78.72	874	$+5.9 / +6.0 / +5.9$
Croston/SBA OR	2,705	1,665	1,040	15,521	16.3	98.30	78.87	803	$+8.2 / +8.4 / +8.3$
Empirical-quantile OR	2,489	1,543	946	13,331	14.8	98.35	78.87	747	$+17.7 / +17.8 / +17.7$
Simulation-tuned (r, Q)	3,266	2,473	793	11,914	12.6	96.05	84.84	1,973	$-10.4 / -10.3 / -10.3^\dagger$
LLM (run 1)	2,928	1,824	1,104	16,256	17.3	98.11	78.87	910	
LLM (run 2)	2,931	1,850	1,081	16,599	17.0	97.89	78.70	968	
LLM (run 3)	2,929	1,841	1,088	16,958	17.0	98.16	78.90	890	
LLM without ERP	3,014	1,902	1,112	16,507	17.3	98.06	78.89	890	$-8.8^\ddagger$ (vs SAP)

[†] Other comparisons of the ablation are in Section 6.4. [‡] Different operating point (mean fill 96.05%, aggregate fill 84.8%); the paired interval of the cost difference includes zero, and with priced shortages this policy is the cheapest arm (Section 6.8).

6.3. Information asymmetry: what the LLM adds beyond the ERP inputs

First, the SLT-informed OR and its ERP-floor variant are the most expensive arms (4,211 and 4,248): the ERP safety lead time inflates their effective lead time and buffers. Against them the LLM is 30–31% cheaper in every run, which mainly measures how heavily ERP inputs over-buffer.

Second, the *history-only OR*, which uses no ERP inputs, costs 2,764 at a fill of 98.14%. The LLM costs 5.9–6.0% more in all three runs (run 1: +164, interval +91 to +240), with fill differences of -0.03 , -0.25 and $+0.01$ points whose intervals include zero, and is cheaper in only 74–77 of 346 pairs. The stronger data-driven arms widen the gap: the LLM costs 8.2–8.4% more than the Croston/SBA OR (2,705, fill 98.30%; run 1: +223, interval +97 to +362; cheaper in 91–99 pairs) and 17.7–17.8% more than the empirical-quantile OR (2,489, fill 98.35%; +439, interval +329 to +557; cheaper in 44–49 pairs; mean fill 0.24 points lower). Part of the LLM’s advantage over ERP-derived parameters therefore reflects weak ERP benchmarks.

The simulation-tuned (r, Q) policy sits at a different operating point: an aggregate fill of 84.8% and the lowest inventory (11,914 units) but a mean fill of 96.05% and 1,973 stockout days, because it is tuned on the pre-cutoff year. The LLM costs 10.3–10.4% less at zero penalty with a 2.1-point higher mean fill, but the interval includes zero ($-2,527$ to $+857$); once shortages are priced the tuned policy is by far the cheapest arm (Section 6.8).

Third, the LLM improves on ERP-derived parameters (11.3–11.4% cheaper at 0.14–0.40 points lower fill; the interval excludes zero only in run 2) but does not reach the best classical baselines, which also yield a policy for every material at lower cost.

6.4. LLM without ERP planning parameters (ablation)

In this ablation the LLM receives the same inputs as in runs 1–3, but the ERP safety stock and safety lead time are blanked in the CSV and marked “not supplied”; the system prompt file is unchanged. It tests whether the LLM anchors on the ERP planning parameters. All 365 pairs returned a valid, storage-compliant policy, and the policies change only slightly beyond

374 ordinary run-to-run variation: the median ratios of safety stock and reorder
 375 point to run 1 are 1.00; the reorder point differs by more than 10% in 54.6%
 376 of pairs (48.3% and 48.0% between ERP-supplied runs) and the safety stock
 377 in 11.0% (9.8% and 9.5%); the mean ratio is 1.04 for safety stock and 1.05 for
 378 the reorder point. In the hard-limit replay the cost is 3,014 at a mean fill of
 379 98.06%, against 2,928–2,931 in runs 1–3. It is 8.8% below SAP-derived pa-
 380 rameters (interval -566 to -61 ; fill -0.24 points, interval including zero) and
 381 28.4–29.1% below the SLT-informed and ERP-floor ORs, but 9.0% above the
 382 history-only OR (interval $+134$ to $+387$; cheaper in 91 of 346 pairs), 11.4%
 383 above the Croston/SBA OR, 21.1% above the empirical-quantile OR and
 384 7.7% below the simulation-tuned policy. By plant the pattern is unchanged
 385 (-10.8% in Plant A, -21.9% in Plant B, $+41.6\%$ in Plant C against $+36.2\%$
 386 in run 1), so the Plant C loss does not stem from reliance on ERP parameters.
 387 The information-asymmetry concern therefore does not explain the LLM’s
 388 advantage over ERP-derived parameters, and the ERP planning fields ap-
 389 pear to be of limited use to the LLM. The ablation withholds only these
 390 two fields; unit cost, holding rate, ordering cost, MOQ, service target and
 391 storage limit remain available, as they are for all reference arms.

392 *6.5. Storage limits along simulated trajectories*

393 All 365 LLM policies satisfy the static rule $SS+Q \leq \text{max_storage_units}$,
 394 yet in the unconstrained simulation post-receipt on-hand stock exceeds the
 395 limit for 183 pairs (52.9%; 178–183 across runs; 10.2% of pair-days) for the
 396 LLM, 198 (57.2%) for SAP-derived and 230 (66.5%) for the SLT-informed
 397 OR, because inventory on order and lead-time demand add to stock above
 398 $SS + Q$. The static check is thus necessary but not sufficient for any
 399 method. Under the hard-limit replay no arm exceeds the limit by more

than one unit; enforcing it raises the LLM mean cost from 2,850 to 2,928
and SAP-derived from 3,184 to 3,305 without changing the ranking.

6.6. Plants and demand regimes

The average hides a strong plant effect (Table 4, hard-limit replay, run 1).
The LLM costs 14.1% less than SAP-derived in Plant A (cheaper in 179 of
215 pairs) and 22.7% less in Plant B (54 of 65), but 36.2% *more* in Plant C
(cheaper in 8 of 66); the Plant C loss is similar in the other full runs (+32.2%,
+31.5%) and persists under every working-day reconstruction in which Plant
C can be scored (Section 6.8). Against the data-driven baselines the LLM
costs more in every plant: 7.2%, 3.6% and 6.7% more than the history-only
OR, 13.1%, 2.3% and 6.2% more than the Croston/SBA OR, and 20.8%,
7.7% and 29.5% more than the empirical-quantile OR in Plants A, B and
C. The SLT-informed and history-only ORs coincide in Plant C because the
ERP safety lead time is zero for all its pairs.

By demand regime, the 33 lumpy materials show the largest reduction
against SAP-derived (−25.4% in run 1) and the 313 smooth materials −9.8%
at a 0.24-point lower mean fill. On lumpy materials the service effect is
inconsistent (mean fill +0.38 points in run 1, −0.29 in run 2), the history-only
and Croston/SBA ORs cost about 9–10% less than the LLM, and only the
empirical-quantile OR is slightly more expensive (1.2%); with 33 materials
this is indicative only. By terciles of the coefficient of variation, the LLM
costs 15.7%, 7.5% and 12.9% less than SAP-derived (lowest to highest), with
a 0.68-point lower mean fill in the middle tercile.

Table 4: Mean simulated cost per pair and mean fill by plant, run 1, hard-limit replay (common-feasible cohort). The SLT-informed and history-only ORs coincide in Plant C.

Plant (n)	LLM cost	SAP-derived cost	SLT-informed OR cost	History-only OR cost	Croston/SBA OR cost	Empirical-quantile OR cost	LLM vs SAP	Cheaper pairs	Fill LLM / SAP (%)
All (348)	2,928	3,305	4,211	2,784	2,785	2,489	−11.4%	211	98.11 / 98.20
A (215)	2,391	2,782	3,779	2,231	2,114	1,980	−14.1%	179	99.18 / 99.27
B (65)	4,916	6,368	7,229	4,746	4,896	4,566	−22.7%	54	96.23 / 96.27
C (66)	2,719	1,937	2,549	2,549	2,561	2,100	+36.2%	8	96.20 / 96.61

423 6.7. Run-to-run stability

424 Across the three runs the aggregate results are very similar: the LLM
 425 cost relative to SAP-derived is -11.4% , -11.3% and -11.4% , and the policy-
 426 value medians are close (reorder point 1,584, 1,582 and 1,466; order quantity
 427 2,737, 2,777 and 2,693; safety stock 683, 688 and 669). Individual policies
 428 vary, however. Between runs 2 and 3 on the 365 pairs, all three values are
 429 identical for only 36 pairs; every policy resolves to the (r, Q) family, and the
 430 median absolute differences are 81 units (reorder point), 6 (order quantity)
 431 and 3 (safety stock). The reorder point differs by more than 10% in about
 432 48% of pairs, and a few pairs vary strongly (one reorder point was 14,882,
 433 14,947 and 29,218 units in the three runs). Aggregate behaviour is repro-
 434 ducible, but one generation per material is a noisy estimate whose individual
 435 values should be reviewed, in line with the documented non-determinism of
 436 LLMs [16].

437 6.8. Sensitivity analyses

438 *Plant C calendar..* Plant C’s working-day flag required reconstruction, so we
 439 re-scored all policies under four alternatives (any-plant, strict-all, exclusion
 440 of Plant C, calendar-day arrivals). Where Plant C can be scored, the LLM
 441 remains 31–35% more expensive than SAP-derived there in the three full
 442 runs, and the cohort-wide advantage is -10.8% to -11.1% ; the strict and
 443 exclusion variants (280 pairs) show -17.0% to -17.6% . The Plant C result
 444 is therefore not an artefact of the repair; a residual risk is that the LLM saw
 445 Plant C demand series built on the repaired calendar.

446 *Shortage penalty..* Table 5 gives mean total cost per pair (holding, order-
 447 ing and shortage cost) in the hard-limit replay for penalties of one and five

448 times the unit cost. Costs are dominated by the shortage term, and the
 449 differences among the non-tuned arms (0.1–3.7%) should be read with cau-
 450 tion. The simulation-tuned policy is by far the cheapest (1,094,043 at one
 451 times the unit cost, about 33% below SAP-derived), because it was selected
 452 to meet the service target on the pre-cutoff history. Among the other arms
 453 SAP-derived parameters are the cheapest, as they carry more inventory and
 454 fewer stockouts. LLM run 1 then costs 2.4% more than SAP-derived at both
 455 multiples, is about as expensive as the Croston/SBA OR (1,673,003 ver-
 456 sus 1,674,925 at one times), and is cheaper than the SLT-informed (1.2%),
 457 ERP-floor, empirical-quantile and history-only (0.8%) ORs. The LLM’s cost
 458 advantage over ERP-derived parameters is therefore specific to the case in
 459 which service shortfalls are not charged.

Table 5: Mean total cost per pair in the hard-limit replay (346 pairs) when lost sales are charged at a multiple of the unit cost.

Arm	1×	5×
SAP-derived	1,634,136	8,157,462
SLT-informed OR	1,694,183	8,454,072
ERP-floor OR	1,694,220	8,454,109
History-only OR	1,687,191	8,424,899
Croston/SBA OR	1,674,925	8,363,806
Empirical-quantile OR	1,693,784	8,458,966
Simulation-tuned (r, Q)	1,094,043	5,457,151
LLM run 1	1,673,003	8,353,306
LLM run 2	1,683,280	8,404,676
LLM run 3	1,671,759	8,347,082
LLM no ERP	1,689,961	8,437,748

460 7. Managerial Insights

461 1. **Position the tool as a reviewed decision support system, not**
 462 **an autonomous engine.** Single generations differ at the material level
 463 (Section 6.7). The gate-simulate-review structure yields a justified can-
 464 didate for every material and leaves the decision to the planner; whether
 465 this helps planners is untested.

- 466 **2. Benchmark against data-driven baselines.** ERP-derived parameters
467 and ERP-informed ORs are weak benchmarks (safety stocks are often
468 stale or zero; safety lead times inflate buffers), so comparing only with
469 ERP overestimates the benefit: the LLM cost 6%, 8% and 18% more than
470 history-only, Croston/SBA and empirical-quantile baselines.
- 471 **3. Expect plant-level differences.** Against ERP-derived parameters the
472 LLM was cheaper in Plants A and B (14–23%) and on 33 lumpy materials
473 (about 25%, mixed service) but not in Plant C; we did not test whether
474 escalation rules would catch such cases.
- 475 **4. Verify storage along the trajectory.** The static rule $SS + Q \leq \text{limit}$
476 cannot protect a warehouse: 53–66% of pairs breach the limit in simula-
477 tion under each of the three arms checked. Enforce a cap in the order-
478 release logic or treat the limit as a design requirement.
- 479 **5. Price shortages explicitly.** Without a shortage penalty the ranking
480 rewards under-stocking; with a penalty of one unit cost the LLM is about
481 2.4% more expensive than SAP-derived. Set service targets and penalties
482 before reading cost comparisons.
- 483 **6. Freeze and log the configuration.** Treat the system prompt as a
484 controlled file, log every call (input hash, time, session, raw response)
485 and re-run before changing the prompt.
- 486 **7. Cost and latency.** Each generation took several minutes, whereas OR
487 baselines take milliseconds; periodic regeneration would support adapt-
488 able planning [84].

489 8. Discussion, Limitations and Research Perspectives

490 *What the evidence shows..* A prompt-configured LLM generates feasible, re-
491 viewable (r, Q) policies for every material; in simulation they cost less than
492 ERP-derived parameters but more than data-driven OR policies. InventOR
493 is therefore not presented as outperforming optimization: its role is a re-
494 viewed candidate generator, and whether that helps planners is a hypothesis
495 the simulation cannot test. Against the ERP-informed arms alone the LLM
496 would look far better (-11.4% against SAP-derived, -30.5% against the
497 SLT-informed OR).

498 *Design principles for LLM-based decision support..* We derive six principles
499 from this application and offer them as hypotheses for design-science work
500 [7]. **DP1, bound the generator:** the LLM is a candidate generator whose
501 prompt fixes the method, constraints and output schema, with no authority
502 to act. **DP2, verify deterministically and along time:** check outputs
503 with code and feasibility along the trajectory, because point checks can pass
504 while constraints are violated over time (Section 6.5). **DP3, show counter-**
505 **factual evidence:** give the reviewer simulated comparisons with credible
506 references. **DP4, control information in evaluation:** compare against
507 baselines with and without the same information and withhold incumbent
508 inputs in an ablation (Sections 6.3, 6.4). **DP5, freeze and log:** treat the
509 prompt as a controlled file, log every call and repeat runs (Section 6.7). **DP6,**
510 **escalate by rule:** define in advance when proposals are trusted, escalated
511 or rejected, to limit complacency [13] (Section 4.3).

512 *Limitations..* (i) The data come from one manufacturer and three plants. (ii)
513 The model identifier and inference settings are operator-declared. (iii) No

514 planner outcomes were measured: the benefits of reviewability, auditability
 515 and planner support and the rules of Section 4.3 are design intentions that
 516 a field study must test. (iv) The simulation uses deterministic arrival at the
 517 rounded mean lead time, ignores observed receipts, corrections and scrap,
 518 starts with an empty pipeline and does not simulate lead-time variability,
 519 which may bias comparisons that depend on safety lead time; we did not
 520 report service-target attainment. (v) The primary results use a zero shortage
 521 penalty; with a positive penalty the ranking changes. Costs are dominated
 522 by a few high-demand materials and the 346 pairs are not independent,
 523 so the bootstrap intervals understate uncertainty. (vi) The (r, Q) family
 524 is single-item and single-echelon. (vii) Individual policies vary across runs,
 525 and Plant C required calendar reconstruction. (viii) We included no deep
 526 reinforcement-learning or neural forecasting baseline, and the ERP-derived
 527 arm (ERP safety stock with our own order quantity) is not an observed
 528 record of practice.

529 *Research perspectives..*

- 530 1. *Ensembling and hybrids.* Aggregating several generations may reduce
 531 individual-policy noise [42], and combining the best data-driven OR policy
 532 with LLM review or regime-specific choice [85] could capture the OR cost
 533 level and the LLM’s flags.
- 534 2. *Comparison with trained methods.* Benchmarks against deep
 535 reinforcement-learning and other trained policies [53, 46, 47] on
 536 identical data and information would clarify the trade-off between setup
 537 effort and policy quality.
- 538 3. *Richer policy families.* (s, S) policies and regime-specific estimators may

539 help lumpy and Plant C pairs [58, 59].

540 4. *Planner-in-the-loop field trials.* Prospective studies should compare plan-
541 ners with and without the aid on matched materials, measuring planner
542 time, override rates and realized service and inventory, and should exam-
543 ine adoption drivers [28, 27, 5].

544 5. *Standards for evaluation.* LLM-based decision support studies should
545 report an incumbent-independent baseline, an ablation withholding in-
546 cumbent inputs, trajectory-level feasibility and run-to-run variation.

547 9. Conclusion

548 We defined InventOR as a human-reviewed decision support system for a
549 recurring production-planning decision, the setting of replenishment param-
550 eters per plant-material pair, and evaluated it against six design requirements
551 on a three-plant industrial dataset. A prompt-configured LLM generated
552 feasible, justified (r, Q) policies for every material without training, and a
553 deterministic gate, a hard-limit simulation and explicit review rules made
554 them checkable. The proposals cost about 11% less than ERP-derived pa-
555 rameters at slightly lower service, but the LLM costs about 6%, 8% and 18%
556 more than history-only, Croston/SBA and empirical-quantile policies; when
557 lost sales are priced at one unit cost or more, ERP-derived parameters are
558 the cheapest of the non-tuned arms and a simulation-tuned policy is cheapest
559 overall. Storage limits that hold in a static check are exceeded along simu-
560 lated trajectories for every method, and the LLM loses to ERP at one plant.
561 The evidence supports InventOR as a checkable candidate generator inside
562 a reviewed process, not as an optimizer that beats data-driven baselines; the

design principles of Section 8 are hypotheses, and field evidence on planner behaviour and realized outcomes is the central open step.

Appendix A. Formal Definitions

Reference calculation. These are the standard textbook calculations [68, 79]. For pair i , let $\bar{d}_i$ and $\sigma_{D,i}$ be the mean and standard deviation of pre-cutoff working-day demand, L_i and $\sigma_{L,i}$ the mean and standard deviation of the lead time in working days, K_i the ordering cost, h_i the annual holding cost per unit (unit cost times holding rate), z_i the standard normal quantile of the service target, and 250 the working days per year. The order quantity is $Q_i = \max\{\text{MOQ}_i, \lceil \sqrt{2K_i \cdot 250\bar{d}_i/h_i} \rceil\}$. The lead-time-demand standard deviation is $\sigma_{LTD,i} = \sqrt{L_i\sigma_{D,i}^2 + \bar{d}_i^2\sigma_{L,i}^2}$, the safety stock $SS_i = \lceil z_i\sigma_{LTD,i} \rceil$ and the reorder point $R_i = \lceil \bar{d}_iL_i^{\text{eff}} + SS_i \rceil$, where L_i^{eff} is the mean lead time for the history-only OR and the mean lead time plus the ERP safety lead time for the SLT-informed OR and the ERP-floor OR. The SAP-derived arm uses the ERP safety stock and the same Q_i . All policies are projected to $SS_i + Q_i \leq \text{max_storage_units}_i$ when $\text{MOQ}_i + SS_i$ fits.

Simulation. On each post-cutoff working day, demand is served from on-hand stock, unmet demand is lost, and an order of size Q_i is placed when the inventory position (on-hand plus on order) is at or below R_i and arrives after the rounded mean lead time in subsequent working observations. Observed receipts, corrections and scrap are ignored. The total cost is $TC_i = h_i\bar{I}_iT/250 + K_iN_i$, where $\bar{I}_i$ is the mean on-hand stock over the T simulated working days and N_i the number of orders. Fill rate is served divided by demanded units. In the hard-limit replay an order is truncated to $\text{max_storage_units}_i$ minus the inventory position, and no order is placed

588 if the truncated quantity is zero.

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597 The author declares no known competing financial interests or personal
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599 paper.

600 **Data and Code Availability**

601 The raw operational data and material-level artifacts cannot be publicly
602 released because they contain commercially sensitive material, supplier
603 and planning information. A confidentiality-screened release package
604 containing analysis code, the anonymized variable schema, aggregate
605 evaluation results, manuscript sources and prompt files is available at
606 github.com/DressPD/inventor_public. A versioned Zenodo archive of
607 the release is available at <https://doi.org/10.5281/zenodo.22262052>.
608 Supplementary File S1 contains the system prompt used for all runs; the
609 prompt file, the runner code and, for each call, a hash of the input, the

610 UTC time, the session identifier and the raw response are in the release.
611 The model identifier and inference settings are not exposed by the platform
612 and are operator-declared (Anthropic Claude Sonnet 4.6).

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619 **Declaration of Generative AI and AI-Assisted Technologies in the** 620 **Writing Process**

621 During manuscript preparation and revision, the author used OpenAI
622 GPT-5.6 and Anthropic Claude (Sonnet 5.5) models through the OpenCode
623 coding assistant for language editing, restructuring, consistency checks, ci-
624 tation verification against Crossref metadata, LaTeX proofing, and writing
625 and executing the analysis scripts that produce the evidence tables. These
626 assistants are distinct from the Claude Sonnet 4.6 deployment evaluated in
627 the study. The author specified the analyses, reviewed and edited all content
628 and results, and takes full responsibility for the content. No generative AI
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